

Disproof of conjecture 00000001678

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Verdict: FALSE

We disprove conjecture 00000001678 as stated. Its first clause — that the difference between the zero forcing number and the path cover number on trees takes values in $\{0, 1\}$ — is not the point of failure. The failure is the “*exactly*” clause: the asserted classification of the difference-one trees. In fact, by the AIM theorem, $Z(T) = P(T)$ for *every* tree T , so the difference is always 0 and the difference-one class is *empty*, while the class of odd-diameter trees admitting a perfect matching is non-empty. The smallest witnesses are the paths $P_2, P_4, P_6, \dots$; the case $T = P_4$ is worked out below and formalised in Lean 4.

1 The conjecture

Quoted verbatim from `conjectures/00000001678.md`:

English. Conjecture: The difference between the zero forcing number and the path cover number on trees takes values in $\{0, 1\}$, with an explicit tree classification (zero-forcing versus path-cover tree classification); and the trees with difference one are exactly the odd-diameter trees admitting a perfect matching.

中文。猜想：零力数 (zero forcing number) 与路径覆盖数之差在树上取值 $\{0, 1\}$ 且树分类显式 (Z - 路径覆盖树分类)；且差为一的树恰为含完美匹配的奇直径树。

2 Definitions

Throughout, $G = (V, E)$ is a finite simple graph and, later, a tree.

Definition 1 (zero forcing number; colour-change rule). Colour every vertex of G either *black* or *white*. The *colour-change rule* is: if a black vertex u has *exactly one* white neighbour v , then v is turned black; one says that u *forces* v . A *zero forcing set* is a set $S \subseteq V$ of vertices such that, starting with exactly the vertices of S black and all others white and repeatedly applying the colour-change rule, eventually every vertex is black. The *zero forcing number* $Z(G)$ is the minimum cardinality of a zero forcing set.

The iteration terminates because the number of black vertices is non-decreasing and bounded by $|V|$; the closure of S is well defined.

Definition 2 (path cover number). A *path cover* of G is a set of pairwise vertex-disjoint paths whose vertex sets cover V . A path cover is *induced* if each path is an induced subgraph of G . The

path cover number $P(G)$ is the minimum number of paths in a path cover of G .

For a tree the two conventions coincide: a path subgraph of a tree is automatically induced, since a tree has no chords (any additional edge between two vertices of the path would close a cycle).

Definition 3 (diameter and perfect matching). The *diameter* $\text{diam}(G)$ is the maximum graph distance between two vertices of G . A *perfect matching* of G is a set of pairwise disjoint edges whose endpoints together cover every vertex of G .

The two classes compared by the conjecture are therefore

$$\begin{aligned}\mathcal{D} &= \{ T \text{ a tree} : Z(T) - P(T) = 1 \}, \\ \mathcal{R} &= \{ T \text{ a tree} : \text{diam}(T) \text{ odd and } T \text{ has a perfect matching} \},\end{aligned}$$

and the “exactly” clause asserts $\mathcal{D} = \mathcal{R}$.

3 The AIM theorem: the difference is always zero

The following is the relevant theorem from the AIM minimum-rank / zero-forcing programme.

Theorem 4 (AIM; Barioli–Fallat–Hogben et al., 2008). *For every tree T , the zero forcing number and the path cover number agree:*

$$Z(T) = P(T).$$

Reference. F. Barioli, S. Fallat, L. Hogben, et al., *Zero forcing sets and the minimum rank of graphs*, Linear Algebra and its Applications **428** (2008) 1628–1648. The inequality $Z(T) \geq P(T)$ holds for every graph, and equality for trees is the content of the cited theorem. $\square$

Corollary 5. *For every tree T , $Z(T) - P(T) = 0$. Hence the difference-one class $\mathcal{D}$ is empty.*

Corollary 6. *The class $\mathcal{R}$ is non-empty: for instance every path P_{2k} has odd diameter $2k - 1$ and the perfect matching $\{v_1v_2, v_3v_4, \dots, v_{2k-1}v_{2k}\}$.*

Since $\mathcal{D} = \emptyset$ while $\mathcal{R} \neq \emptyset$ by the two corollaries, $\mathcal{D} = \mathcal{R}$ is impossible. The remainder of the paper makes this concrete by evaluating a single smallest witness.

4 The witness $T = P_4$

Let P_4 have vertices v_1, v_2, v_3, v_4 and edges v_1v_2, v_2v_3, v_3v_4 .

Lemma 7 ($Z(P_4) = 1$).

Proof. Upper bound. Take the zero forcing set $S = \{v_1\}$. Vertex v_1 is black and has the unique white neighbour v_2 , so v_1 forces v_2 . Now v_2 is black; its neighbours are v_1 (black) and v_3 (white), so v_2 forces v_3 . Now v_3 is black; its neighbours are v_2 (black) and v_4 (white), so v_3 forces v_4 . All four vertices are black. Thus $Z(P_4) \leq 1$.

Lower bound. The empty set has no black vertex, so the colour-change rule never applies and no vertex is ever forced; since $|V| = 4 > 0$ the empty set is not a zero forcing set. Hence $Z(P_4) \geq 1$. Therefore $Z(P_4) = 1$. $\square$

The explicit forcing sequence is

$$\{v_1\} \xrightarrow{v_1 \rightarrow v_2} \{v_1, v_2\} \xrightarrow{v_2 \rightarrow v_3} \{v_1, v_2, v_3\} \xrightarrow{v_3 \rightarrow v_4} \{v_1, v_2, v_3, v_4\}.$$

Lemma 8 ($P(P_4) = 1$).

Proof. Upper bound. The single path $v_1v_2v_3v_4$ covers all four vertices and is a subgraph of P_4 ; as noted above, in a tree every path subgraph is induced, so it is an admissible (induced) path cover with one path. Hence $P(P_4) \leq 1$.

Lower bound. Every path cover has at least one path because $V \neq \emptyset$. Hence $P(P_4) \geq 1$. Therefore $P(P_4) = 1$. $\square$

Theorem 9 (the difference on P_4 is zero).

$$Z(P_4) - P(P_4) = 1 - 1 = 0 \neq 1.$$

In particular $P_4 \notin \mathcal{D}$.

Proof. Immediate from Lemma 7 and Lemma 8. $\square$

Lemma 10 (P_4 is an odd-diameter tree with a perfect matching). $\text{diam}(P_4) = 3$, which is odd, and $\{v_1v_2, v_3v_4\}$ is a perfect matching of P_4 . Consequently $P_4 \in \mathcal{R}$.

Proof. Along a path the graph distance is the index difference, so $\text{dist}(v_i, v_j) = |i - j|$. The largest value is $\text{dist}(v_1, v_4) = 3$; thus $\text{diam}(P_4) = 3$, an odd number. The edges v_1v_2 and v_3v_4 are disjoint and their endpoints $\{v_1, v_2\}$ and $\{v_3, v_4\}$ cover all four vertices, so they form a perfect matching. $\square$

Theorem 11 (the “exactly” clause is false). Conjecture 00000001678 is false as stated. Indeed

$$P_4 \in \mathcal{R} \quad \text{but} \quad P_4 \notin \mathcal{D},$$

so $\mathcal{D} \neq \mathcal{R}$ and the assertion that the difference-one trees are exactly the odd-diameter trees admitting a perfect matching fails.

Proof. Lemma 10 gives $P_4 \in \mathcal{R}$; Theorem 9 gives $P_4 \notin \mathcal{D}$. $\square$

Remark 12 (even smaller and infinite witnesses). P_2 is an even smaller counterexample: $Z(P_2) = P(P_2) = 1$, its diameter is 1 (odd), and its unique edge is a perfect matching, yet the difference is 0. More generally, for every $k \geq 1$ the path P_{2k} has diameter $2k - 1$ (odd), the perfect matching $\{v_1v_2, \dots, v_{2k-1}v_{2k}\}$, and $Z = P = 1$, so

$$P_{2k} \in \mathcal{R} \setminus \mathcal{D} \quad \text{for every } k \geq 1.$$

The witnesses are therefore infinite in number.

Remark 13 (what remains true). The first clause of the conjecture, that $Z - P \in \{0, 1\}$ on trees, is compatible with the AIM theorem ($Z - P = 0$ always), so that clause alone is not refuted. What is refuted is the classification of the difference-one trees, which is vacuous.

5 Computational corroboration

The accompanying program `reproduce.py` (Python 3, standard library only) brute-forces Z over all subsets with the colour-change closure and P by exhaustive path-cover search. It verifies the table for P_n , $n = 1, \dots, 10$:

n	$Z(P_n)$	$P(P_n)$	$Z - P$	diam	odd?	perfect matching?
1	1	1	0	0	no	no
2	1	1	0	1	yes	yes
3	1	1	0	2	no	no
4	1	1	0	3	yes	yes
5	1	1	0	4	no	no
6	1	1	0	5	yes	yes
7	1	1	0	6	no	no
8	1	1	0	7	yes	yes
9	1	1	0	8	no	no
10	1	1	0	9	yes	yes

In particular every odd-diameter P_n admitting a perfect matching ($n = 2, 4, 6, 8, 10$) has $Z - P = 0$. Enumerating all trees on $1, \dots, 10$ vertices (the 201 isomorphism classes; Prüfer sequences for $n \leq 8$, canonical rooted trees for $n = 9, 10$, cross-validated) gives:

trees with $Z - P = 1$	0
trees with odd diameter and a perfect matching	14
trees with $Z = P$ (AIM)	201

The difference-one class is empty, while the class of odd-diameter trees admitting a perfect matching is not. The script prints **PASS** and exits with status 0 if and only if these verifications succeed.

6 Formalisation

The witness P_4 is formalised in Lean 4 in the accompanying `lean4/` project (core Lean only, `import Std`, no `Mathlib`, no `sorry`). In it, vertices are of type `Fin 4`, adjacency is $i \text{---} (i + 1)$, the colour-change closure is a computable function of the colouring, the zero forcing number is a minimum over all 16 subsets, and the path cover number is a minimum over all 256 idempotent vertex maps encoding partitions. The main theorems are

```

theorem singleton_zero_forces : isZFS (single 0) = true
theorem zeroForcingNumber_P4 : zeroForcingNumber = 1
theorem pathCoverNumber_P4 : pathCoverNumber = 1
theorem diameter_P4 : diameter = 3
theorem diameter_P4_odd : isOdd diameter = true
theorem P4_has_perfect_matching : hasPerfectMatching = true
theorem difference_P4 : zMinusP = 0
theorem difference_P4_ne_one : zMinusP ≠ 1
theorem conjecture_00000001678_refuted : ...

```

Every theorem is discharged by computation (`decide`); `Check.lean` prints `#print axioms` for them and shows that no proof uses `sorryAx` (only `propext`, from the standard library, appears).

7 Files

File	Description
README.md	Verdict, conjecture quote, definitions, the P_4 computation, the P_{2k} family, the AIM context, file list, reproduction commands, submission status.
main.tex	This LaTeX source (standalone <code>article</code> , builds with <code>tectonic main.tex</code>).
build/main.pdf	The compiled document.
reproduce.py	Python 3 brute force: Z by colour-change closure, P by exhaustive path covers, the P_n table, and all trees on $1, \dots, 10$ vertices.
lean4/lean-toolchain	Pins <code>leanprover/lean4:v4.33.1</code> .
lean4/lakefile.toml	Lake configuration; library <code>Main</code> , no dependencies.
lean4/Main.lean	Lean 4 formalisation of the witness (core Lean, no <code>sorry</code>).
lean4/Check.lean	<code>#print axioms</code> audit of every theorem.
lean4/README.md	Statement table, proof strategy, scope note.

8 Reproducing

```
$ python3 reproduce.py
...
PASS: P_4 has Z = 1, P = 1, Z - P = 0, odd diameter 3 and a
perfect matching
$ tectonic main.tex
$ cd lean4 && lake build
$ cd lean4 && lake env lean Check.lean
```

The numerical checks are dependency-free. The Lean project uses the toolchain pinned in its `lean-toolchain` file and imports only `Std`.

9 Status against the submission rules

The submission rules require the LaTeX source code, a PDF document, and a Lean 4 project.

- **LaTeX source** — present (`main.tex`), a standalone `article` using only `geometry`, `amsmath`, `amssymb`, `amsthm`, `array`, and `parskip`; it compiles with `tectonic main.tex` without external figures or non-standard packages.
- **PDF document** — `build/main.pdf`, produced by `tectonic`; it is a rendered copy of `main.tex` and exists non-empty in this submission.
- **Lean 4 project** — present under `lean4/` with `lean-toolchain`, `lakefile.toml`, `Main.lean`, and `Check.lean`. `lake build` exits 0, and `lake env lean Check.lean` prints the axioms of each theorem with no `sorryAx` and no `Mathlib` dependency. The formalisation establishes, by finite computation, that $Z(P_4) = 1$, $P(P_4) = 1$, $\text{diam}(P_4) = 3$ is odd, P_4 has a perfect matching, and $Z(P_4) - P(P_4) = 0 \neq 1$.